IRIS FLOWER CLUSTERING PROJECT

Mini Project 3 — K-Means Clustering and PCA

Student Name: _____
Course / Branch: _____
Submitted To: _____

1. Objective

Apply K-Means with k=3 to the Iris dataset, visualize clusters, compare them with known species labels, and use PCA to reduce four dimensions to two.

2. Dataset

The Iris dataset is included with Scikit-learn and contains 150 samples, four numeric measurements, and three known species. Labels are used only for comparison after unsupervised clustering.

3. Tools

Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, and Jupyter Notebook.

4. Methodology

• Load the built-in Iris dataset.

• Standardize numeric features using StandardScaler.

• Train K-Means using three clusters.

• Use PCA to project the four features into two components.

• Plot K-Means clusters and true species labels.

• Compare labels using a crosstab and Adjusted Rand Index.

• Evaluate cluster separation using the silhouette score.

5. Evaluation

Silhouette Score: indicates how well-separated the clusters are; higher is generally better.

Adjusted Rand Index (ARI): measures agreement between clusters and known labels while adjusting for chance. Cluster IDs are arbitrary.

PCA explained variance: reports the share of variance retained by the two plotted components.

6. Results (fill after running)

Silhouette Score: _____

Adjusted Rand Index: _____

Variance explained by two PCA components: _____

Attach both plots generated by the notebook/script.

7. Conclusion

This project demonstrates unsupervised clustering and dimensionality reduction. K-Means groups samples by feature similarity, while PCA enables two-dimensional visualization. The crosstab and ARI show how clusters align with known species. Run the notebook to obtain actual metrics.

8. Viva Questions

• What is unsupervised learning?

• How does K-Means work?

- Why is k set to 3?
- Why standardize features?
- What is PCA?
- Why are cluster IDs arbitrary?
- What does silhouette score measure?
- What does ARI measure?